MSG No. 105.218 P_I4_2mc [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'mm'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,4,5,6,10,11,15,16]`
2	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[2,3,7,8,9,12,13,14]`

Table 2: Wyckoff site: `4b`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[0, 0, z+\frac{1}{2}\right]$	`[9,12,13,14]`
4	$[0, 0, z]$	`[10,11,15,16]`

Table 3: Wyckoff site: `8c`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x, x+\frac{1}{2}, z\right]$	`[1,16]`
2	$\left[\frac{1}{2}-x, x, z+\frac{1}{2}\right]$	`[2,13]`
3	$\left[x+\frac{1}{2}, -x, z+\frac{1}{2}\right]$	`[3,14]`
4	$\left[-x, \frac{1}{2}-x, z\right]$	`[4,15]`
5	$\left[-x, x+\frac{1}{2}, z\right]$	`[5,10]`
6	$\left[x, \frac{1}{2}-x, z\right]$	`[6,11]`
7	$\left[\frac{1}{2}-x, -x, z+\frac{1}{2}\right]$	`[7,12]`
8	$\left[x+\frac{1}{2}, x, z+\frac{1}{2}\right]$	`[8,9]`

Table 4: Wyckoff site: `8d`, site symmetry: `.m.`

No.	position	mapping
1	$[x, 0, z]$	`[1,6]`
2	$\left[0, x, z+\frac{1}{2}\right]$	`[2,8]`
3	$\left[0, -x, z+\frac{1}{2}\right]$	`[3,7]`
4	$[-x, 0, z]$	`[4,5]`
5	$\left[x+\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[\frac{1}{2}, x+\frac{1}{2}, z\right]$	`[10,16]`
7	$\left[\frac{1}{2}, \frac{1}{2}-x, z\right]$	`[11,15]`
8	$\left[\frac{1}{2}-x, \frac{1}{2}, z+\frac{1}{2}\right]$	`[12,13]`

Table 5: Wyckoff site: `16e`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$\left[-y, x, z+\frac{1}{2}\right]$	`[2]`

continued ...

Table 5

No.	position	mapping
3	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[-x,\, y,\, z]$	[5]
6	$[x,\, -y,\, z]$	[6]
7	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[7]
8	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$	[10]
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[13]
14	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[16]